Ive Botunac_

AI Research Engineer · Assistant Professor · PhD in AI
ivebotunac.com info@techprimo.ai linkedin.com/in/ivebotunac Google Scholar
orcid.org/0000-0003-3561-7089 github.com/ivebotunac x.com/ivebotunac
Zadar, Croatia



AI research engineer, assistant professor and founder. I build agentic AI systems, lead partnerships, and publish – from LLM-driven trading systems to a mental-health app with 1.5 million+ downloads.

18+	10+	10	40
years in software development	years in AI	published papers	course certificates

01 ABOUT

AI researcher and engineer with more than 18 years in software development, over ten of them in AI. I help companies adopt AI from strategy to implementation: preparing organizations, building agentic AI systems, and teaching teams to work with AI. My background combines entrepreneurship, leadership and research. I founded and ran my own ventures, led AI and R&D teams, and work on European defence projects with partners across Europe. Today I lead partnerships at ALFATEC Group and CroAI, contribute to national AI policy, and teach computer science as an assistant professor.

02 EXPERIENCE

ALFATEC Group
Full-time · Zagreb, Croatia — 2020 – now · 5 yrs 11 mos

Director of Partnerships — Jul 2025 – Present
Lead strategic partnerships and acquisitions for the group. On European Defence Fund projects I handle partner communication, divide the AI work and coordinate teams across international consortia, among them HYBRID (€3.3M, an AI reconnaissance drone) and LODESTAR II (€7.1M, AR and AI for soldier systems). I also run partner relations on national EU projects with AI components.

Director of Research & Development — Dec 2021 – Aug 2025
Led the R&D department and its AI work: projects with international partners, machine-learning models for client and in-house products, and the MLOps practice that kept them running. Also shaped hiring and built the teams behind the in-house products.

Data Scientist — Dec 2020 – Nov 2021
Built machine-learning solutions tailored to client needs, and led the forming of a development team for the company's own products.

University of Zadar
Part-time · Zadar, Croatia — 2026 – now · 4 mos

Assistant Professor — Jul 2026 – Present
Teaching computer science at the newly founded Department of Mechatronics and Robotics, from its first generation of students. I help build the study programme and bring industry practice and AI research into the classroom.

05 SKILLS

AI AGENTS & LLM APPS
LangGraph · LangChain · MCP · Claude Code · OpenAI · Gemini · Anthropic · AWS Bedrock AgentCore · ChromaDB · AssemblyAI

ML RESEARCH
PyTorch · Stable-Baselines3 · Transformers · PEFT · Unsloth · TRL · Optuna · scikit-learn · pandas · TensorFlow · Backtrader

MOBILE
Ionic Vue · Capacitor · RevenueCat · Swift · ARKit · LiDAR

WEB
Nuxt · Vue · TypeScript · Pinia · Tailwind · React · Redux Toolkit · TanStack

BACKEND & CLOUD
Node.js · Firebase · Google Cloud · AWS · Express · DynamoDB · SQS · CDK · PostgreSQL · Redis · Docker

ENGINEERING
Vitest · Playwright · GitHub Actions · Jest · Cypress · MLOps

06 CERTIFICATIONS

2025 Agentic AI
DeepLearning.AI · Course

2023 Generative AI with Large Language Models
Coursera · Course

2023 Machine Learning Specialization
Coursera · 3 courses

2022 Product Owner Training (Certified Product Owner)
Product Manager HQ · Training

2022 Natural Language Processing Specialization
Coursera · 4 courses

2021 Reinforcement Learning Specialization
Coursera · 4 courses

Stockifi
Contract · Oslo, Norway
2025 – now · 1 yr 1 mo

AI Research Engineer — Oct 2025 – Present
Research and build agentic AI for a hospitality platform trusted by 500+ restaurants across Europe, from bars to hotel F&B, that gives them full visibility over their food costs. My focus is invoice automation: AI agents that read supplier invoices line by line and match every line to the venue's own items, so recipe costs follow each supplier price change.

Martian & Machine
Contract · Berlin, Germany
2025 – now · 1 yr 2 mos

AI Research Engineer — Sep 2025 – Present
Research and build agentic AI at Martian & Machine, an agentic AI company that helps enterprises move from AI pilots to an AI workforce. I work with the team on autonomous coding agent systems for a large enterprise client, taking on real engineering work inside its codebases.

CroAI – Croatian AI Association
Volunteer · Zagreb, Croatia
2024 – now · 1 yr 11 mos

Board President — Dec 2025 – Present
Leading CroAI into its next chapter after a year as acting President: a new statute, the Management and Supervisory Boards, and the direction of Croatia's AI community.

Vice President (acting President) — Dec 2024 – Dec 2025
Helped grow Croatia's AI landscape from 200 to 500+ actors, connecting companies, startups, academia and government. 3,600+ educators trained on ethical AI through “AI se educiraj”, 80+ Heroes workshops for students, and a seat in the working group for Croatia's AI Strategy 2026–2032.

Primo Technology
Self-employed · Zadar, Croatia
2025 – now · 1 yr 9 mos

Founder — Feb 2025 – Present
My indie dev studio. I design, build and look after a portfolio of 13 focused AI-powered apps on the App Store and Google Play, among them Primo Notes, Primo Scale and Primo Nautic, together with their websites, all run through an agentic AI development workflow of my own.

Sintelly
Full-time · Zadar, Croatia
2017 – 2025 · 8 yrs 8 mos

Founder & CEO — May 2017 – Dec 2025
Founded and led Sintelly, a mobile app that pairs AI with cognitive behavioural therapy and reached over 1.5M downloads worldwide. Built and shipped the AI-driven CBT tools that guide users through therapy techniques, making mental-health support available to anyone with a smartphone.

ZadarConnect
Volunteer · Zadar, Croatia
2022 – 2025 · 2 yrs 10 mos

President — Sep 2022 – Jun 2025
Led ZadarConnect, a non-profit that brings together everyone in Zadar interested in tech and business. Ran a series of meetups and co-organised the SheepAI Hackathon (88 participants, €35,000 prize pool) to grow the city's tech community.

Ministry of Defence (Croatian Air Force)
Full-time · Zadar, Croatia
2009 – 2017 · 8 yrs 1 mo

Technician — Feb 2009 – Feb 2017

2021	Data Engineering, Big Data & ML on GCP	Coursera · 5 courses
2021	Google Cloud: Cloud Data Engineer (prep)	Coursera · 6 courses
2019	DeepLearning.AI TensorFlow Developer	Coursera · 4 courses
2019	Advanced Machine Learning on Google Cloud	Coursera · 5 courses
2018	Machine Learning on Google Cloud	Coursera · 5 courses
2018	Deep Learning Specialization	Coursera · 5 courses
2016	Machine Learning	Coursera · Course

03 EDUCATION

PhD, Informatics
University of Rijeka, Faculty of Informatics and Digital Technologies · 2018 – 2025
An Automated Stock Market Trading System Based on Deep Reinforcement Learning

MSc, Management
University of Zadar, Department of Economics · 2016 – 2018
Applying machine learning methods to predict the movement of the stock market index

BSc, Nautical Studies & Maritime Transport Technology
University of Zadar, Maritime Department · 2013 – 2016
Evolution of the cyber threat in the maritime industry

04 PUBLICATIONS

2025 **Automated Trading Framework Using LLM Driven Features and Deep Reinforcement Learning**
Ive Botunac, Tomislav Petković, Jurica Bosna

2025 **Implementation of a Multi-Agent AI System for Financial Trading Decision Making**
Ive Botunac · *Oeconomica Jadertina*

2024 **Comparing Fine-Tuning and Prompt Engineering for Multi-Class Classification in Hospitality Review Analysis**
Ive Botunac, Marija Brkić Bakarić, Maja Matetić · *Applied Sciences (CC)*

2024 **Opportunities of Gen AI in the Banking Industry with regards to the AI Act, GDPR, Data Act and DORA**
Ive Botunac, Natalija Parlov, Jurica Bosna · *13th Mediterranean Conference on Embedded Computing (MECO), SCOPUS*

2024 **Optimization of Traditional Stock Market Strategies Using the LSTM Hybrid Approach**
Ive Botunac, Jurica Bosna, Maja Matetić · *Information 15.3 (2024): 136, SCI index* · doi.org/10.3390/info15030136

2024 **Artificial Intelligence, ESG Innovations in Tourism, and Personal Data Management**
Natalija Parlov, Nada Bašić, **Ive Botunac** · *Acta Economica Et Turistica*

2020 **The Effect of Feature Selection on the Performance of LSTM Neural Networks in Stock Market Predictions**
Ive Botunac, Ante Panjkota, Maja Matetić · *31st DAAAM International Symposium, SCOPUS*

2019 **The Importance of Time Series Data Filtering for Predicting Stock Market Movement Using Neural Networks**
Ive Botunac, Ante Panjkota, Maja Matetić · *30th DAAAM International Symposium, SCOPUS*

2017 **Analysis of the Security Challenge in Maritime Cloud Computing**
Ive Botunac, Juraj Poljak, Dino Županović · *7th International Maritime Science Conference, Split (TRID)*

2017 **Analysis of Software Threats to the Automatic Identification System**
Ive Botunac, Marijan Gržan · *Brodogradnja 68(1): 97–105, SCI index*